

Expectation



Reality



1a. Introduction

Jin Hong

jin.hong@uwa.edu.au

1a. Introduction: outline

- ◇ Team
- ◇ Location
- ◇ Schedule
- ◇ Assessments
- ◇ Remember

Team



Jin Hong
Unit coordinator
Room CS1.10
jin.hong@uwa.edu.au



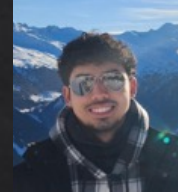
Larry Huynh
Senior Lab Facilitator



Lab:
Alian Haidar
Lab facilitator



Lab:
Jack Sun
Lab facilitator



Lab: Thu
Muslim Gilani
Lab facilitator



Teaching Operations (team)
Admin team
Room Main reception
teachingops-team2@uwa.edu.au

Emergency

- ◆ General emergency: call campus security 24/7 at (08) 6488 2222
- ◆ In super emergency: call emergency at 000
- ◆ In all buildings, we have an emergency procedure such as this picture ->
 - ◆ Please take a time and read it.
- ◆ For more details, please have a read through our emergency procedure for various potential incidents
 - ◆ <https://www.uwa.edu.au/about-us/location-and-campus/campus-services/safety/hazards-incidents-and-emergencies/emergency-procedures>

Emergency procedures

For life threatening emergencies dial 000 (if phoning from UWA dial 0 for outside line if using office phone; if using TEAMS no need to dial 0 first). Also dial UWA Security on [\(+618\) 6488 2222](tel:+61864882222).

Code RED - Fire/smoke	▼
Code PURPLE - Bomb threat	▼
Code BLUE - Medical emergency	▼
Code BLACK - Personal threat	▼
Code YELLOW - Internal (UWA) emergency	▼
Code BROWN - External emergency	▼
Code ORANGE - Evacuation	▼

Location

- ◆ Lecture

- ◆ Venue: Austin Lecture Theatre (ARTS159)
- ◆ Time: Mondays 1pm - 3pm

- ◆ Labs

- ◆ Lab 1: EZONE CENT 210 (Mondays 3 - 5pm)
- ◆ Lab 2: EZONE CENT 209 (Tuesdays 1 - 3pm)
- ◆ Lab 3: CSSE 201 (Thursdays 12 - 2pm)

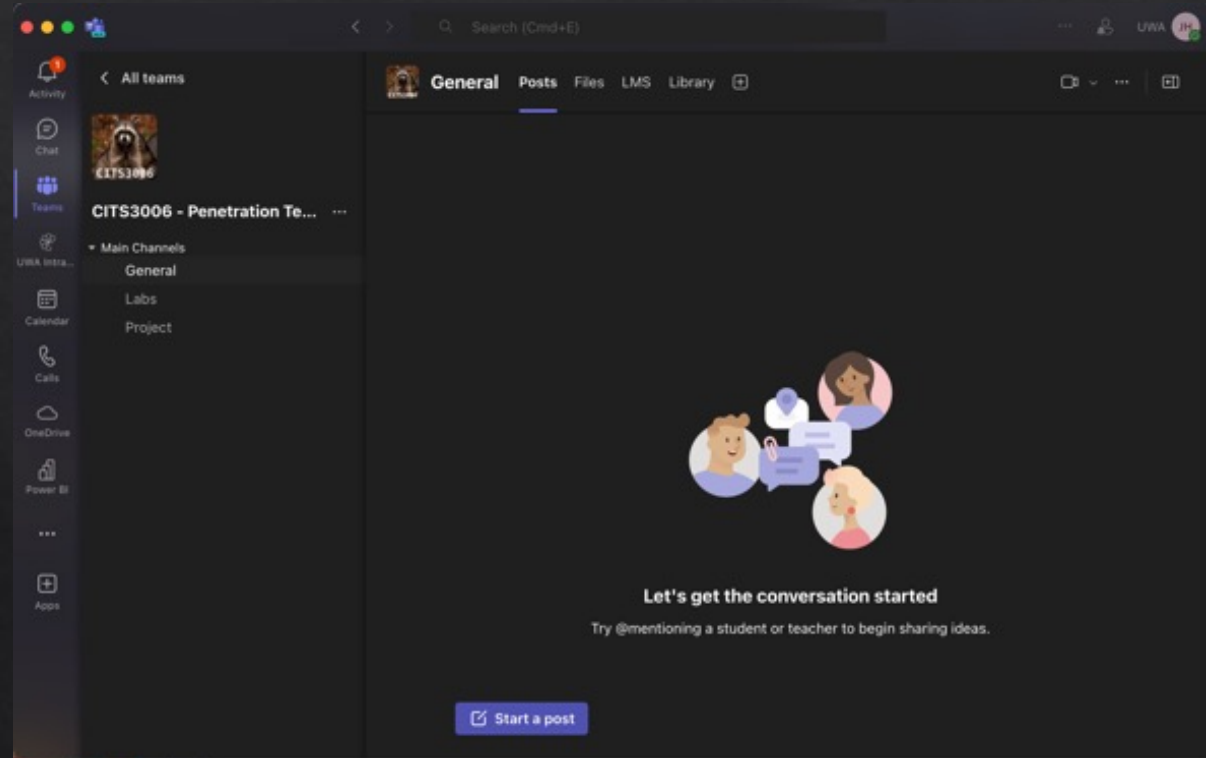
- ◆ Consultation

- ◆ Office at Computer Science building room 1.10

Changes will be announced (if any)

Location

- ◆ Feel free to ask questions on teams anytime
 - ◆ Someone will answer you in their next availability
- ◆ Other students feel free to join the discussion



Course overview: Term 3

Labs start first week!

Week	Lecture	Lab	Assessments
1	Ethics + Reconnaissance	Lab 0: Setup and Linux	
2	Network Exploits + Malware	Lab 1: Network Exploits	
3	More Malware	Lab 2: Malware	
4	Reverse Engineering	Lab 3: Reverse Engineering	Lab Quiz 1
5	Rowhammer (TBC)		Lab Quiz 1 Demo
6	Privilege Escalation	Lab 4: Privilege Escalation	

Course overview: Term 4

Week	Lecture	Lab	Assessments
7	Command and Control		Lab Quiz 2
8	Web Security	Lab 5: Web Security	Project out
9	TBC (is a public holiday)	Lab 6: Active Directory	
10	Active Directory		Lab Quiz 3
11	Maybe OSINT?		Lab Quiz 3 Demo
12	Guest Lectures and/or Revision		Project demo

Assessments

Assessment Item	When	Covers	Worth (total)
Lab Quizzes	Weeks 4, 7, 10	Materials up to the assessment week (excluding), and also excluding previously assessed weeks.	60% (20% each)
Group Project	Week 8	Various	40%
Optional research project	over the semester	Various	prepares you for honours/masters research

Please note, all dates are tentative and subject to change!

How does the lab quiz work?

- ◆ We will be using two formats for the lab quizzes:
 - ◆ Complete given tasks over the set amount of time, and do the demo later (the following week lab)
 - ◆ Complete given tasks and demo during the scheduled lab
- ◆ For the latter method, you will be supervised by a lab facilitator to complete a timed lab quiz (~60 mins but can vary).
- ◆ You cannot receive mark if your attendance to the demo is not confirmed by the lab facilitator for both format.
 - ◆ i.e., you must be supervised /invigilated by the lab facilitator to receive marks.
 - ◆ Attendance is required for all F2F students (unless approved preivously e.g., special consideration).

How does the lab quiz work?

- ◇ For the demo, we will be using a peer-marking system, where you will be formed into a group of 4 (or thereabout), and each person will have ~15 mins to do the demo.
- ◇ Marking keys will be provided for peer markers to use and submit the marking.
- ◇ The peer markers will also provide feedback to the demonstrator including the mark range.
- ◇ Facilitators will be available for any moderations and review of the peer marking process.
- ◇ All submitted marks will be moderated based on the report submitted, and finalised marks will be uploaded to csmarks.

Group Project

- ◆ Details to be confirmed but this year I am thinking:
 - ◆ You are in a group (self-assignments)
 - ◆ Part 1 (50%) : You will configure a vulnerable VM
 - ◆ Part 2 (30%) : You will exploit another group's VMs
 - ◆ Part 3 (20%): You will harden a randomly assigned VM (and demo Part 2)
 - ◆ Part 4 (steal points): You will exploit any hardened VMs
 - ◆ You will submit accompanying reports (group and individual)
 - ◆ Each part will be 1 week long.
- ◆ OR
 - ◆ You are in a group (self-assignments)
 - ◆ Exploit prepared VMs live (hackathon/ctf style).
 - ◆ One day or the weekend to complete.
 - ◆ Submit report and do the demo.
 - ◆ But TBC.

(Optional) Research project

- ◇ You don't have to do this – purely optional for enthusiasts
- ◇ If interested, contact me directly via email with the header:
"CITS3006 research project"
- ◇ You will be assigned in a group of 3-4 students, and a mentor to assist with the research.
- ◇ This will be a good practice for your honours research or Masters research project (if you are in BACS or thinking of doing Masters).
- ◇ Spaces will be limited for up to 2 groups, so a placement is not guaranteed.
 - ◇ Preference will be given to Honours students if we have too many applications.
- ◇ We'll have regular meetings (weekly) to discuss about the research project and make progress throughout the year.
- ◇ Goal is to publish a paper with accompanying software/tool.
 - ◇ If accepted to a conference, I can send you to present our work (expenses covered).

Prerequisite

- Prerequisites: CITS2002 or CITS2005
- Recommended: CITS2006 and CITS3002
- This unit requires the student to be knowledgeable in various aspects of CS.
- This unit expects students to be knowledgeable in various programming languages such as Java, Python and C (or be able to quickly catch up as necessary).
- You should also be familiar with basic discrete mathematics and number theory.
- I highly recommend taking this unit in third year after taking various CITS units (including ones listed above) for preparations.

Please note

- This unit is/does NOT:
 - cover a comprehensive penetration testing techniques
 - Pentesting and cybersecurity is a broad discipline!
 - about all of the latest and greatest attacks
 - Covering selected but important fundamental themes
 - Read online sources instead to keep up to date
 - cover ethical, legal or economic issues
 - We will touch on ethical issues briefly later, but not focus on them

Please note

- Lectures provide **theoretical** and **conceptual** understandings of the topics presented
 - But we will also do some practicals in the lectures
 - So, you should bring your laptop if you want to try them in class as well
- Labs and project provide **practical skills** of the topics presented
- The contents in Lectures, labs and project will be related, but they are all **INDEPENDENT** learning materials
 - i.e., you will be learning new things in each lecture, lab and project.

FAQ

- ◆ Why is lab quiz timed?
 - ◆ Being able to solve problems in a reasonable time is a key ability in any roles. We have more flexible assessment – group project that runs over multiple weeks.
 - ◆ Also, assessment contents are set such that it is reasonable to solve within the given timeframe.
 - ◆ Having said that, I will explore some alternative options for this (as mentioned earlier).

- ◆ I was told to use chatGPT to prepare for lab quizzes, is this true? why?
 - ◆ This is a misconception. You are welcome to use chatGPT in all assessments in CITS3006 to your best ability, but you'll find it quickly that all tasks given are not fully solvable by chatGPT (and alike). We encourage using any and every tool you'll find yourself helpful /useful, including chatGPT, to support your learning, as well as in assessments as it seem fit.
 - ◆ We found some students solve all assessments without the use of chatGPT, while some benefited using chatGPT for certain tasks. It is totally up to you if you wish to use it or not.

FAQ

- ◇ I feel like the instructions in the assessments are too vague, why?
 - ◇ It leaves sufficient room for exploration. Also, in any software development/testing environments, you will never have perfect requirements. It is up to you to consult with the client to check that the requirements are met or make reasonable justifications that would be acceptable by the client.
 - ◇ Same in the assessments, you are also told to discuss with me if any confusions arise, and reasonable and practical justifications and assumptions are accepted for design choices of your solutions.

FAQ

- ◇ Why don't we learn about tool X and tool Y?
 - ◇ I understand some tools are the preferred tool in certain industries. But the goal of the unit is not to train you to use those tools, but focus on the fundamental aspects of the topics covered.
 - ◇ 10 years ago, none of those tools existed (or most).
 - ◇ 10 years later, you probably be using some other tools.
 - ◇ However, the fundamentals don't change that quickly
 - ◇ E.g., thinking about algorithms, data structures etc.

Any other questions?

Copyright notice

Commonwealth of Australia

Copyright Regulations 1969

WARNING

Materials in this unit (CITS3006 – Penetration Testing) have been reproduced and communicated to you by or on behalf of The University of Western Australia pursuant to Part VB of the *Copyright Act* 1968 (**the Act**).

The material in this communication may be subject to copyright under the Act. Any further reproduction or communication of this material by you may be the subject of copyright protection under the Act.